4×4 Mesh NoC – Synthesis Results (Table 3)

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1. Setup

Synopsys Design Compiler U-2022.12, SAED14 nm RVT NLDM slow corner (SS, 0.72 V, 125 °C), `compile_ultra`, input and output delays 40% of the clock period. The three designs use the same TCL and SDC; only `rtl/mkNoc.v` differs. All three were run at a 1.0 ns clock (1.0 GHz) and met timing there: total negative slack 0.00 ns, no design-rule violations. The critical path in each is register to register (f2f).

2. Table 3

Design	Feature	Closing frequency (slack ≥ 0)	Cell count (leaf)	Design area (μm^2)
1	Round-robin arbiter	1.0 GHz (1.0 ns), slack 0.00 ns	239 929	562 978.02
2	Fixed, transit first (best)	1.0 GHz (1.0 ns), slack 0.06 ns	226 142	481 332.37
3	Fixed, local first	1.0 GHz (1.0 ns), slack 0.00 ns	220 172	486 393.44

Design	Levels of logic (f2f)	Critical path	Cell area	Sequential cells	Buf/inv cells
1	17	0.98 ns	141 862	69 048	35 994
2	20	0.91 ns	134 496	68 864	30 109
3	13	0.97 ns	131 488	68 780	25 696

3. Throughput per area

Design	Measured B_N	B_N at 1.0 GHz	B_N / design area
1	183.79 B/cycle	183.79 GB/s	326.5 kB/(s·μm^2)
2	191.21 B/cycle	191.21 GB/s	**397.3** kB/(s·μm^2)
3	187.50 B/cycle	187.50 GB/s	385.5 kB/(s·μm^2)

4. Notes

Only the 1.0 ns period is reported, so the closing frequency is the one actually met, not a limit. Design 2 has 0.06 ns of slack left and may close at a shorter period. Design 1 is the largest: the pointer update and rotating search add logic (17% over Design 2). Buffer storage, and so almost all sequential cells, is the same in all three. The QoR reports are submitted as `design_<n>_qor.rpt`, with the exact TCL and SDC for each run.